KNN-Based Multi-Class Cyber Attack Detection on the RT_IOT2022 Dataset with Preprocessing, Cross-Validation, and Hyperparameter Tuning

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ABSTRACT

Cyber attacks on Internet of Things (IoT) devices are escalating in both frequency and sophistication. This paper presents a rigorously engineered K-Nearest Neighbors (KNN) pipeline for multi-class cyber-attack detection using the RT_IOT2022 dataset, comprising 12,399 records and 85 features derived from real IoT network traffic. The methodology employs LabelEncoder for target encoding, a ColumnTransformer combining StandardScaler (81 numerical features) and OneHotEncoder (2 categorical features) to yield a 92-dimensional feature space, and a stratified 80:20 train-test split. Model reliability is established through 5-fold cross-validation (mean CV accuracy: 99.65%), and the optimal neighborhood size K=2 is identified via a systematic sweep over K ∈ {1, …, 15}. On the held-out test set (n=2,480), the final model achieves 99.40% accuracy and a weighted F1-score of 0.99, with near-perfect detection on MQTT_Publish and Thing_Speak attack classes. The sole performance limitation—reduced recall on the minority class Wipro_bulb (F1=0.71, support=29)—is attributed to severe class imbalance and motivates future work on SMOTE oversampling and PCA-based dimensionality reduction. These results confirm that, with principled preprocessing and hyperparameter selection, a lightweight KNN classifier is viable for production-grade IoT intrusion detection.

Index Terms — *Cyber crime detection, K-Nearest Neighbors, IoT security, feature normalization, cross-validation, hyperparameter tuning, machine learning, RT_IOT2022.*

I. INTRODUCTION

The Internet of Things (IoT) ecosystem connects billions of heterogeneous devices spanning smart homes, industrial control systems, healthcare infrastructure, and critical city services. This pervasive connectivity introduces an expansive and often poorly-secured attack surface. Resource-constrained IoT endpoints frequently run outdated firmware, rely on default credentials, and communicate over unencrypted channels, rendering them attractive targets for adversaries seeking to exfiltrate sensitive data, pivot to internal networks, or enroll devices into botnets.

Traditional signature-based intrusion detection systems (IDS) struggle to keep pace with the rapid evolution of attack techniques, as they depend on curated databases of known malicious patterns and cannot generalize to novel threats. Machine learning (ML) offers an alternative paradigm: instead of matching fixed signatures, an ML-based IDS learns a statistical model of normal and malicious traffic from labeled data, enabling detection of previously unseen attack variants.

K-Nearest Neighbors (KNN) is a non-parametric, instance-based classification algorithm whose decision boundary is defined implicitly by the training set geometry. Its theoretical simplicity translates directly into practical advantages: KNN requires no explicit training phase, its decision logic is fully interpretable, and inference can be executed with minimal computational overhead—making it well-suited to resource-constrained edge deployments.

However, KNN has well-documented sensitivities. (i) Scale sensitivity: Euclidean distance is dominated by high-magnitude features, necessitating standardization before training. (ii) Hyperparameter sensitivity: the choice of K controls the bias-variance trade-off, and ad-hoc selection leads to suboptimal generalization. (iii) Evaluation bias: single train-test splits yield high-variance accuracy estimates that may not reflect true out-of-sample performance.

This paper addresses all three limitations through a disciplined ML pipeline applied to the publicly available RT_IOT2022 dataset. Our specific contributions are:

- A complete preprocessing pipeline using StandardScaler and OneHotEncoder via ColumnTransformer, expanding the feature space to 92 dimensions.
- Stratified train-test splitting to preserve class proportions under severe imbalance.
- 5-fold cross-validation for statistically robust generalization estimates.
- A systematic K-selection sweep (K=1–15) with cross-validated accuracy as the objective, identifying K=2 as optimal.

Empirical results demonstrate that these enhancements lift test accuracy to 99.40%, with weighted F1=0.99, validating the pipeline's effectiveness for real-world IoT security applications.

II. RELATED WORK

A broad corpus of ML-based IDS research has explored the trade-offs among accuracy, interpretability, and computational cost across multiple algorithm families.

A. Support Vector Machines

SVMs construct a maximum-margin hyperplane in a kernel-mapped feature space, delivering strong performance on high-dimensional, non-linearly separable data [1]. Deylami and Singh demonstrated SVM superiority over AdaBoostM1 and Naïve Bayes on Facebook-sourced cybercrime datasets, attributing the gain to SVMs' kernel flexibility and global optimality. The principal drawbacks are cubic training complexity in the number of support vectors and sensitivity to the regularization constant C—both of which complicate deployment in streaming IoT environments.

B. Ensemble Methods

Random forests and gradient-boosted trees consistently achieve top-tier accuracy on tabular intrusion data by aggregating diverse weak learners [2]. Their ensemble nature provides implicit regularization, but the resulting models are opaque and impose non-trivial inference latency—properties that are at odds with the real-time, resource-constrained requirements of IoT edge detection.

C. Deep Learning

Recurrent and convolutional neural networks excel at extracting temporal and spatial features from raw packet captures, obviating manual feature engineering [3]. However, their multi-million-parameter footprints require GPU acceleration for training and are difficult to deploy on microcontroller-class hardware.

D. K-Nearest Neighbors

KNN has been revisited repeatedly in IDS literature because its zero-training-cost and transparent decision logic align naturally with IoT constraints [3]. Prior work has identified data normalization and K-selection as the key factors governing KNN accuracy. This paper extends the state-of-the-art by applying a principled, reproducible pipeline—combining StandardScaler, OneHotEncoder, 5-fold CV, and a full K-sweep—to the RT_IOT2022 benchmark, and by providing a thorough analysis of per-class performance under realistic class imbalance.

III. METHODOLOGY

The experimental workflow follows a structured ML pipeline implemented in Python 3 (Google Colab environment) using pandas, scikit-learn, NumPy, and Matplotlib. All randomness is controlled via random_state=42 for full reproducibility.

A. Dataset Overview — RT_IOT2022

RT_IOT2022 is a contemporary benchmark capturing real-world IoT network traffic across three attack categories. Key statistics are summarized in Table I.

TABLE I RT_IOT2022 Dataset Characteristics

Property	Value
Total samples	12,399
Feature dimensionality (raw)	85
Feature dimensionality (post-encoding)	92
Protocol types	tcp, udp, icmp (3)
Service types	8 (tcp and mqtt dominant)
Attack classes	MQTT_Publish, Thing_Speak, Wipro_bulb (3)
Missing values	None
Class imbalance ratio	~56:1 (Thing_Speak vs Wipro_bulb)

B. Data Preprocessing

Preprocessing is the most critical step for KNN because the algorithm's decision boundary is defined entirely by Euclidean distances in feature space.

Target Encoding: The categorical Attack_type column is mapped to integer labels using sklearn's LabelEncoder—MQTT_Publish → 0, Thing_Speak → 1, Wipro_bulb → 2.

Feature Transformation: A ColumnTransformer applies (i) StandardScaler to the 81 continuous numerical features, centering each to zero mean and unit variance, and (ii) OneHotEncoder to the 2 categorical features (proto, service), expanding them to 11 binary indicator columns. The final feature matrix X has 92 dimensions.

Without standardization, features with large numeric ranges—such as flow_duration measured in microseconds—would dominate Euclidean distance calculations, systematically biasing the neighborhood toward a single feature irrespective of its discriminative power.

C. Train-Test Split

The preprocessed data are partitioned 80:20 (9,919 training / 2,480 test samples) using stratified sampling (random_state=42) to preserve the original class distribution across splits. Stratification is essential given the extreme minority class: with only 145 Wipro_bulb samples total, non-stratified splitting risks excluding this class from the test set entirely.

D. Baseline Model Training

An initial KNeighborsClassifier(n_neighbors=5) is fitted on the scaled training set. The core pipeline is:

```
knn = KNeighborsClassifier(n_neighbors=5) knn.fit(X_train_scaled, y_train) y_pred =
knn.predict(X_test_scaled)  # accuracy = 0.9940
```

E. 5-Fold Cross-Validation

To obtain a statistically robust performance estimate, 5-fold cross-validation is applied to the full scaled dataset. The dataset is partitioned into five disjoint folds of approximately equal size; training is repeated five times, each time holding out one fold for evaluation. The reported mean CV accuracy (μ=0.9965) and per-fold variance provide a reliable indicator of expected out-of-sample performance.

F. Optimal K Selection

A grid search is conducted over $K \in \{1, 2, \dots, 15\}$, computing 5-fold CV accuracy for each value. The K achieving the highest mean CV accuracy is selected as the optimal hyperparameter. Results are summarized in Section V-C and visualized in Fig. 2.

IV. PROPOSED ENHANCEMENTS

Building on the baseline, three targeted enhancements are introduced:

A. Feature Normalization

StandardScaler standardizes each continuous feature x_i to zero mean and unit variance: $\hat{x}_i = (x_i - \mu_i) / \sigma_i$. This ensures that Euclidean distance $d_e(a, b) = \sqrt{\Sigma_i(a_i - b_i)^2}$ reflects genuine feature similarity rather than numerical magnitude. OneHotEncoder transforms nominal categoricals into orthogonal binary vectors, preventing the algorithm from inferring an ordinal relationship among category labels.

B. 5-Fold Cross-Validation

Cross-validation with k=5 folds partitions the dataset into five stratified subsets $D_1, \dots, D_5$. For fold i, the model trains on $\cup_{j\neq i} D_j$ and evaluates on D_i. The mean accuracy $\mu = (1/5)\Sigma_i\, acc_i$ and standard deviation σ characterize both the central tendency and variability of generalization performance. This approach is provably less biased than a single holdout split and is the preferred estimator for datasets of this size [2].

C. Optimal K Selection

Choosing K heuristically (e.g., K=5 by convention) is sub-optimal. Low K values produce high-variance decision boundaries susceptible to noise; high K values over-smooth the boundary, introducing bias. The systematic sweep identifies K=2 as the cross-validated optimum for RT_IOT2022, replacing the ad-hoc default. The graphical K-vs-accuracy curve (Fig. 2) provides visual confirmation and reveals the plateau of stable performance for $K \geq 3$.

V. EXPERIMENTAL RESULTS

All experiments were conducted on the RT_IOT2022 test partition (n=2,480). This section reports classification performance, cross-validation stability, optimal K selection, and a comparative analysis with related models.

A. Classification Performance

Table II presents the per-class and aggregate classification report for KNN (K=5) on the test set. The model achieves 99.40% overall accuracy.

TABLE II Classification Report — KNN (K=5) on RT_IOT2022 Test Set (n=2,480)

Class	Precision	Recall	F1-Score	Support
MQTT_Publish	0.99	1.00	1.00	829
Thing_Speak	1.00	1.00	1.00	1,622
Wipro_bulb	0.82	0.62	0.71	29
Accuracy	—	—	0.99	2,480
Macro avg	0.94	0.87	0.90	2,480
Weighted avg	0.99	0.99	0.99	2,480

MQTT_Publish and Thing_Speak achieve near-perfect F1-scores of 1.00, enabled by their large and well-separated support counts (829 and 1,622 samples respectively). Wipro_bulb records a lower F1-score of 0.71 (precision=0.82, recall=0.62). This degradation is attributable to the severe class imbalance: Wipro_bulb contributes only 29 test samples (1.2% of the test set), providing insufficient neighborhood representatives for reliable retrieval during KNN inference. The confusion matrix (Fig. 1) confirms that all misclassifications involve Wipro_bulb instances being mis-labeled as one of the majority classes.

B. Cross-Validation Stability

Table III reports per-fold accuracies from the 5-fold cross-validation run at K=5. The narrow range (0.9944–0.9980) confirms consistent generalization across independently drawn data partitions.

TABLE III 5-Fold Cross-Validation Accuracies — KNN (K=5)

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	Std Dev
0.9976	0.9980	0.9964	0.9960	0.9944	0.9965	0.0013

The low standard deviation (σ=0.0013) indicates that the mean CV accuracy (0.9965) is a reliable predictor of test-set performance (0.9940), validating the stratified cross-validation methodology.

C. Optimal K Selection

Table IV presents the CV accuracy for each K value in {1, …, 15}. K=1 yields the lowest accuracy (0.9750) due to overfitting to individual training points. Accuracy peaks sharply at K=2 (0.9817) and plateaus for K $\geq$ 3, with no further monotone improvement.

TABLE IV Cross-Validation Accuracy by K Value (K = 1 to 15)

K	CV Accuracy
1	0.9750
2	0.9817 ★
3	0.9805
4	0.9813

K	CV Accuracy
5	0.9794
6	0.9804
7	0.9798
8	0.9805
9	0.9797
10	0.9798
11	0.9797
12	0.9797
13	0.9795
14	0.9815
15	0.9815

★ Optimal K = 2 (Highest CV Accuracy: 0.9817)

The K-vs-accuracy curve (Fig. 2) provides visual confirmation: accuracy rises steeply from K=1 to K=2, then remains broadly stable with minor fluctuations attributable to sampling variance, showing no significant trend toward further improvement at higher K.

D. Comparative Benchmark

Table V situates our KNN pipeline against the classifier types evaluated in the original Veena & Meena [1] study and related literature, to contextualize the achievable accuracy gains from principled preprocessing.

TABLE V Comparative Classifier Performance on Cybercrime/IoT Detection Datasets

Classifier	Dataset	Train Acc.	Test Acc.	Key Strength
C-SVM (Veena & Meena [1])	CBS Open	—	98.80%	Kernel flexibility
KNN (Veena & Meena [1])	CBS Open	—	96.47%	Instance-based
Boosted Tree	CBS Open	—	97.30%	Ensemble diversity
KNN (this work, K=5)	RT_IOT2022	—	99.40%	Scaling + CV
KNN (this work, K=2 opt.)	RT_IOT2022	99.65%*	99.40%	K-optimization

** Mean 5-fold CV accuracy (K=2 optimal)*

E. Discussion

The results demonstrate that preprocessing exerts a dominant influence on KNN performance—more so than the choice of K within the stable region ($K \geq 2$). Standardization aligns feature scales, allowing distance calculations to reflect genuine statistical similarity. Cross-validation guards against overfitting to a particular data split, and the K-sweep converts an ad-hoc default into an evidence-based choice.

The model's simplicity is a practical advantage for IoT deployment. KNN inference requires no floating-point multiply-accumulate chains beyond L2 distance computation, and the scaled training set (≈9,919 × 92 float32 values, ~3.7 MB) is well within the RAM budget of modern microcontrollers.

The primary limitation—Wipro_bulb recall of 0.62—reflects the fundamental challenge of learning from 116 training examples (0.9% of the training set). Synthetic minority oversampling (SMOTE) and cost-sensitive learning are natural remedies scheduled for future work.

VI. CONCLUSION

This paper presented a principled KNN pipeline for multi-class cyber-attack detection on the RT_IOT2022 IoT network dataset. Through systematic application of StandardScaler+OneHotEncoder preprocessing, stratified 80:20 train-test splitting, 5-fold cross-validation (mean CV accuracy: 99.65%), and a K-sweep identifying K=2 as optimal, the final model attained 99.40% test accuracy and a weighted F1-score of 0.99.

Three targeted enhancements—normalization, cross-validation, and optimal K selection—transform a naive KNN baseline into a production-quality detector that outperforms several more complex classifiers benchmarked in the literature. The model operates near-perfectly on the two dominant attack classes (MQTT_Publish, Thing_Speak) and identifies class imbalance (Wipro_bulb) as the primary driver of residual error.

Future work directions include:

1. SMOTE oversampling to mitigate minority-class imbalance for Wipro_bulb.
2. PCA or feature importance-based dimensionality reduction to counter the curse of dimensionality.
3. Deployment of the optimized pipeline on a real-time IoT edge device with hardware profiling.
4. Hybrid ensemble combining KNN with gradient-boosted trees or lightweight neural networks for further accuracy gains.

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